

MEHDI CHEBBI

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EDUCATION

HIGHER INSTITUTE OF COMPUTER SCIENCE, UNIVERSITY OF TUNIS EL MANAR (ISI) June 2022
Bachelor's Degree in Embedded Systems

PRIVATE HIGHER SCHOOL OF ENGINEERING AND TECHNOLOGY (ESPRIT) September 2025
Engineering Degree, Specialization in IT Architecture & Cloud Computing

INTERNSHIPS

INTERNSHIP AT THE SAHARA AND SAHEL OBSERVATORY July 2024 – September 2024

- Developed an automated system for downloading Sentinel-1, Sentinel-2, and Sentinel-3 satellite images using Bash scripts.
- Established a data center to efficiently store and manage the satellite data. Created a web application for visualizing and analyzing various indices (e.g., NDVI, NDWI) and layers, with integrated data download features.
- Successfully deployed the system on cloud infrastructure to ensure scalability and robust performance.

END-OF-STUDIES INTERNSHIP AT THE SAHARA AND SAHEL OBSERVATORY February 2025 – June 2025

- Built a secure, production-grade private cloud infrastructure across two physical machines using Kubernetes, with Calico for networking, MetallLB for load balancing, and NGINX ingress for traffic control.
- Deployed Prometheus and Grafana for observability, with RBAC enforcing access control. Integrated Falco and Falcosidekick for runtime threat detection, along with a custom webhook and network policies to automatically quarantine suspicious pods.
- Set up Horizontal Pod Autoscaler (HPA) for resource-efficient scaling and installed Cockpit for VM management. Implemented GlusterFS for distributed storage across nodes.
- Developed and hardened a CI/CD pipeline to build, test, scan, and deploy frontend and backend applications, securing secrets and validating artifacts before release. Also created a Telegram bot for periodic cluster status updates and real-time /status queries to improve operational visibility.

WORK EXPERIENCE

CLOUD & SYSTEMS INTERN (PART-TIME STUDENT) September 2024 – September 2025

THE SAHARA AND SAHEL OBSERVATORY

- Developed and deployed a robust Drupal-based platform for the W-Arly-Pendjari (WAP) Complex, a critical cross-border conservation initiative spanning Benin, Burkina Faso, and Niger.
- Managed the entire application lifecycle end-to-end, with a strong emphasis on infrastructure management.
- Provisioned and configured the production Linux server, implemented stringent security measures to safeguard the environment against common threats, and maintained continuous, reliable operation post-deployment to support ongoing conservation efforts.

INFRASTRUCTURE ARCHITECT & DEVOPS ENGINEER
THE SAHARA AND SAHEL OBSERVATORY

September 2025 – Present

- Defined and led the migration from a single-application-per-server model to a bare-metal Kubernetes platform, improving availability, scalability, and infrastructure utilization.
- Architected and operated a production Kubernetes cluster, including networking and traffic management with Calico, MetallLB, and Ingress controllers.
- Deployed, maintained, and supported multiple production applications, implementing horizontal pod autoscaling (HPA) and zero-downtime deployments for improved reliability.
- Built and maintained CI/CD pipelines with Jenkins, automating deployments, security checks, and code quality validation, reducing deployment cycles from 30 minutes to 1 minute.
- Deployed Longhorn for distributed, highly available persistent storage..
- Implemented end-to-end observability with ELK, Prometheus, and Grafana, and created multi-channel health monitoring bots (Telegram, Email, Facebook) for real-time alerts.
- Enforced Kubernetes security and secret management, including RBAC, runtime threat detection with Falco and Falco Sidekick, automated pod quarantine via webhooks, strict NetworkPolicies, and HashiCorp Vault for secrets management.
- Achieved ~99.98% uptime, delivering a stable and highly available production platform.

PERSONAL PROJECTS

INTEGRATED CLOUD PROJECT

- Designed and deployed an end-to-end private cloud platform using OpenStack (Nova, Neutron) with Etcd for distributed configuration and state management
- Implemented and optimized a Kubernetes cluster on top of the cloud infrastructure, integrating Prometheus and Grafana for observability and performance monitoring
- Built and deployed a cloud-native full-stack application (Angular / Spring Boot), running seamlessly on the Kubernetes environment

ZERO-TRUST SECURITY INFRASTRUCTURE

- Designed and deployed an end-to-end private cloud platform using OpenStack (Nova, Neutron) with Etcd for distributed configuration and state management
- Deployed HashiCorp Vault to securely manage 50+ secrets, enabling dynamic credentials and automated rotation
- Integrated Falco for runtime threat detection with custom rules and automated response workflows
- Built a container security scanning pipeline using Trivy and Snyk, integrating vulnerability detection into CI/CD

KUBEMATE (AI POWERED KUBERNETES ASSISTANT | KUBEMATE.DEV)

- Developed a read-only AI-powered Kubernetes assistant to diagnose cluster and workload issues by inspecting live cluster state and resource metadata
- Enabled natural-language debugging of Kubernetes resources, allowing users to query failed pods and receive precise root-cause explanations
- Implemented cluster topology visualizations, including:
 - node-level topology with health and connectivity status
 - pod-to-node and service topology with real-time status mapping
- Built a container security scanning pipeline using Trivy and Snyk, integrating vulnerability detection into CI/CD
- Built an interactive cluster browser with namespace-based navigation, providing:
 - on-demand pod log inspection
 - interactive terminal access within pods
 - in-pod file system exploration via a web-based interface
- Designed the system with strict read-only permissions, ensuring safe introspection without risk to production workloads

AUTOMATED CI/CD PIPELINE SETUP

- Built an end-to-end CI/CD pipeline using Jenkins with GitHub webhooks to trigger automated builds on commit
- Integrated SonarQube for static code analysis and quality gates
- Automated Docker image build and publishing to Docker Hub, with Nexus for artifact management
- Implemented automated testing using JUnit and Mockito

ADDITIONAL

- **Technical Skills:**

Linux server administration, Docker, Kubernetes, Helm, Prometheus, Grafana, ELK Stack, Bash scripting, Python, Jenkins, CI/CD, GitOps, HashiCorp Vault, Falco, Calico, MetalLB, NGINX Ingress, Longhorn, OpenStack

- **Languages:**

Native Arabic; Fluent in English ; Proficient in French ;